

# Jiwoo Kim

PH.D. CANDIDATE · QUANTITATIVE METHODS IN PSYCHOLOGY

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## Education

### Cornell University

Ithaca, U.S.

#### PH.D. CANDIDATE IN PSYCHOLOGY

08. 2023 - current

- Major in Psychology, Minor in Applied Statistics
- Advisor: Dr. Felix Thoemmes
- Dissertation topic: Causal Inference with Missing and Carelessly Generated Data

### Seoul National University

Seoul, Republic of Korea

#### MA IN EDUCATION

03. 2020 - 02. 2022

- Major in Educational Measurement and Evaluation
- Advisor: Dr. Sun-Geun Baek
- GPA: 4.3/4.3
- Dean's list for every semester (four times)
- Master's thesis title: The Effect of Self-Directedness on College Adjustment through Academic Resilience

### Ewha Womans University

Seoul, Republic of Korea

#### BA IN EDUCATION AND PSYCHOLOGY

03. 2016 - 02. 2020

- Major in Education and Psychology
- Undergrad Research Advisor: Dr. Kyoung-Hye Seo
- GPA: 4.13/4.3 Summa cum laude
- Dean's list for every semester (eight times)

## Research Interests

### CAUSAL INFERENCE

causal mediation analysis; causal graphical model; missing data; careless responding; quasi-experimental

### STRUCTURAL EQUATION MODEL

latent class analysis; factor analysis

## Publications

### PUBLISHED

**Kim, J.**, & Thoemmes, F. J. 2026. Treatment effects. *Encyclopedia of Measurement in Social Sciences* (2nd ed.).  
<https://doi.org/10.1016/B978-0-443-26629-4.00243-4>

**Kim, J.**, & Thoemmes, F. J. 2025. Causal assumptions of the two-wave longitudinal mediation model. *Multivariate Behavioral Research*. <https://doi-org/10.1080/00273171.2025.2443360> (Abstract)

**Kim, J.**, Baek, J. W., Lee, H. S (co-first author)., & Choi, Y. J (second author). 2025. Development and validation of the teacher competency diagnostic scale in online learning environment for preservice teacher using scenario method. *SAGE Open*, 15(2). <https://doi-org/10.1177/21582440251344753>

Baek, S. G., **Kim, J.** (corresponding author), & Park, H. 2021. Development and validation of the academic resilience scale for undergraduate students using scenario method. *The SNU Journal of Education Research*, 30(3), 29-57.

## IN PRESS

**Kim, J., & Thoemmes, F. J. Systematic Literature Review for Mediation Analysis** *Collabra: Psychology*

- ▶ Systematically analyze current trends of conducting mediation analysis in social psychology
- ▶ Suggest appropriate ways for mediation analysis
- ▶ Preregistration and materials available at <https://osf.io/dtk4h/overview>

## UNDER REVIEW

**Kim, J., & Thoemmes, F. J. A Causal Approach to Careless Responding**

- ▶ Reframe careless responding as a causal problem rather than treating it solely as a form of measurement error

**Thoemmes, F. J. & Kim, J. The Roles of Machine Learning and Large Language Models in Causal Inference**

- ▶ Examined the potential uses of machine learning and large language models across three stages of causal inference: design, analysis, and auditing.

**Kim, J. & Kim, Y. Statistical Evidence of Sex Discrimination Using Instrumental Inequality**

- ▶ Defined the notion of discrimination using DAG
- ▶ Proved whether there is statistical evidence of discrimination utilizing instrumental variable on a discrimination case

**Soleimani, A. D., Kim, J. & Sternberg, R. J. An Augmented Triangular Theory of the Expression of Love**

- ▶ Using LPA, distinguished different love expression latent profiles

## IN PROGRESS

**Kim, J. Partial Identification with Careless Responding**

- ▶ Propose a causal partial-identification approach to careless responding, deriving bounds for target estimands under alternative assumptions about how careless responses are generated.

**Kim, J., & Thoemmes, F. J. To Remove or Not to Remove? The Consequences of Careless Responding in Psychological Data**

- ▶ Provide simulation-based guidance on when careless responses should be eliminated, conditional on the causal structure of careless responding

**Kim, J., & Thoemmes, F. J. Graphical Criteria of Recoverability under Not Missing at Random Case**

- ▶ Distinguish subtypes of missing-not-at-random according to their bias occurrence using a causal graphical model

**Kim, J., & Thoemmes, F. J. Causal Graphical Representation of Difference-in-Discontinuity**

- ▶ Developing a graphical representation of the difference-in-discontinuities design to clarify the causal assumptions required for valid application

**Kim, J., & Thoemmes, F. J. Causal Assumptions for Two-Wave Longitudinal Mediation Analysis**

- ▶ Confirm existing assumptions for two-wave longitudinal mediation using causal graphical model

**Kim, J., & Thoemmes, F. J. On the Importance of Taking Causal Assumptions in Developmental Psychopathology Seriously.**

- ▶ Invited contribution to a special issue of *Development and Psychopathology*. Draft submitted.

**Thoemmes, F. J., & Kim, J. Review of *Causal Mediation Analysis* by Geoffrey T. Wodtke and Xiang Zhou.**

- ▶ Invited book review for *The American Statistician*. In preparation.

## Presentations

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\* *presenting author*

### ORAL PRESENTATIONS

\***Kim, J., & Kim, Y. 2026. Credible Evidence of Sex Discrimination Using Instrumental Inequality (Forthcoming).** Center for the Study of Inequality. Ithaca, U.S.

\***Kim, J., & Thoemmes, F. 2026. Causal Approach for Careless Responding .** Symposia Presentation: International Meeting of the Psychometric Society (IMPS). Seoul, South Korea.

**Kim, J., & \*Thoemmes, F. 2026. A Systematic Literature Review of Causal Mediation Analysis in Social Psychology.** Oral Presentation: EASP General Meeting 2026. Strasbourg, France.

- \***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: Michigan State University. East Lansing, U.S.
- \***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: International Conference of Educational Research (ICER). Seoul, South Korea
- \***Kim, J.**, & Thoemmes, F. 2025. Systematic Literature Review for Mediation Analysis . Oral Presentation: Brown Bags. Ithaca, U.S.
- \***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Paper Presentation: 2025 International Conference on Education Research (ICER). Seoul, South Korea
- Kim, J.** , Baek, J., & \*Lee, H. 2025. Development of the Pre-service Teacher Competency Scale in an Online Learning Environment. Oral Presentation: American Educational Research Association (AERA), Denver, U.S.
- \*Kim, Y., & **Kim, J.** 2022. Statistical Evidence for Inequality: Applying Causal Restriction of Instrumental Variable. Oral Presentation: Korea Inequality Research Network Conference, Seoul, Korea.

## POSTER PRESENTATIONS

- \***Kim, J.**, & Thoemmes, F. 2026. Revisiting Sequential Ignorability in the Presence of Missing Data (Forthcoming). Poster: 2026 The Society of Multivariate Experimental Psychology (SMEP). Durango, U.S.
- \***Kim, J.**, & Thoemmes, F. 2026. Causal Approach for Careless Responding . Poster: 2026 American Causal Inference Conference (ACIC). Salt Lake City, U.S.
- \***Kim, J.**, & Thoemmes, F. 2025. Graphical Criteria of Recoverability under Not Missing at Random Case . Poster: 2025 American Causal Inference Conference (ACIC). Detroit, U.S.
- \***Kim, J.**, & Thoemmes, F. 2024. Causal Assumptions of the Two-Wave Longitudinal Mediation Model . Poster: 2024 The Society of Multivariate Experimental Psychology (SMEP). Ithaca, U.S.
- \***Kim, J.**, & Kim, Y. 2024. Credible Evidence of Gender Discrimination Using Instrumental Inequality. Poster: 2024 American Causal Inference Conference (ACIC). Seattle, U.S.
- \***Kim, J.** 2022. Table 2 Fallacy in Latent Class Analysis. Poster: 22nd International Conference on Educational Research (ICER). Seoul, Korea.
- Baek, J., \***Kim, J.**, Lee, H., Choi, Y.& Kim, S. 2022. Pre-Service Teachers' Experiences and Perceptions of Teacher Preparation Programs: How Are Teachers Educated to Teach Virtually?. Poster: 22nd International Conference on Education Research (ICER). Seoul, Korea.
- \***Kim, J.**, & Park, H. 2022. Measurement and Analysis of S University students' academic resilience. Online Poster: SNU BK21 Seminar
- \***Kim, J.**, Nam, C., & Seo, Y. 2017. A Narrative Exploration of the EBS Internet Lecture Experience of Low-Income High School Students. Poster: Korea Educational Research Society, Daejeon, Korea.

## Teaching Experience

### Human Bonding

*Cornell University*

#### TEACHING ASSISTANT

*Spring 2026*

- Supported instruction for more than 800 students in a large-enrollment undergraduate course on interpersonal relationships and human behavior.
- Responded to student questions, facilitated discussions, and provided guidance on course readings and assignments.

### Statistics and Research Design

*Cornell University*

#### TEACHING ASSISTANT

*Fall 2025*

- Supported students in implementing and interpreting analyses in R, connecting statistical output to the underlying research question.
- Assisted students in understanding basic statistics, including NHST and the Bayesian approach.

### Statistics and Research Design

*Cornell University*

#### GUEST LECTURER

*Fall 2025*

- Introduced how to use directed acyclic graphs and explained confounders, mediators, and colliders and their roles.

## Quantitative Methods-2

Cornell University

### GUEST LECTURER

Spring 2025

- Introduced the counterfactual framework for causal mediation analysis, including the definitions and interpretation of natural direct, natural indirect, and controlled direct effects.

## Causal Mediation Analysis

GESIS Workshop Series

### TEACHING ASSISTANT

Fall 2024

- Supported participants in implementing causal mediation analyses in R and connecting the code to the underlying estimands and assumptions.
- Provided targeted methodological references in response to participants' questions, enabling further study beyond the workshop.
- Participant feedback described my support as "very active and helpful" and "very attentive," specifically noting the provision of relevant references in real time.

## Multiple Regression Course (Center for Teaching and Learning)

Seoul National University

### TEACHING ASSISTANT

Spring 2021

- Guided students in implementing and interpreting multiple regression models in R.

## Moral Education

Sahmyook Middle School

### PRACTICING TEACHER

Spring 2019

- Designed flipped-learning activities that used class time for discussion and application of moral and ethical reasoning.
- Provided individualized guidance to middle-school students during discussion-based activities.

## Mathematics

Samsung Dream Class

### AFTERSCHOOL TEACHER

Spring 2017 – Fall 2018

- Taught mathematics to underprivileged middle school students.
- Delivered a special lecture on college life and career pathways.

## Korean for North Korean Defectors

Yeomyung School

### AFTERSCHOOL TEACHER

Spring 2018

- Prepared students for the Korean fluency test.
- Advised a delinquent student as a homeroom teacher.
- Organized a career lecture featuring professionals from various fields.

## Professional Experience

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### The Korea Chamber of Commerce and Industry

Seoul, South Korea

#### SENIOR RESEARCHER

Aug. 2022 – Mar. 2023

- Developed National Competency Standard test items for vocational high school students.
- Applied the Rasch model to analyze test results.
- Equated tests using the bookmark method.
- Wrote policy reports on vocational education and student competency evaluation.
- Collaborated with the Ministry of Education on test development.
- Revised tests to reflect changes in the national curriculum.

### Korea Institute for Research in the Behavioral Sciences

Seoul, South Korea

#### RESEARCHER

Jan. 2022 – Jul. 2022

- Developed test items to assess competency and morality.
- Revised and updated norms for psychological assessments.
- Provided individualized feedback to test takers.
- Analyzed personality profiles using the CPI test scale.
- Assessed predictive validity and reported results to client firms.

### Brain Korea 21, Department of Education, Seoul National University

Seoul, South Korea

#### RESEARCHER

Sep. 2021 – Jan. 2022

- Assessed academic resilience among undergraduate students using psychometric scales.
- Supported development and validation of new measurement tools.
- Participated in educational methodology workshops.

## Awards, Fellowships, & Grants

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|------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2026 | <b>The Society of Multivariate Experimental Psychology (SMEP)</b> , Travel grant                                                           | \$2,000  |
| 2026 | <b>Woowa Small Business Owners' Children Scholarship</b> , Woowa Brothers                                                                  | \$10,000 |
| 2026 | <b>Psychometric Society Travel Award (IMPS)</b> , ACS Ventures sponsorship                                                                 | \$1,000  |
| 2024 | <b>Appenzeller Memorial Fellowship</b> , Ewha Foundation                                                                                   | \$3,000  |
| 2023 | <b>Sage Fellowship</b> , Cornell University                                                                                                | \$64,186 |
| 2021 | <b>Scholarship</b> , Korea Association of University Women<br><b>Excellence RA Award</b> , Seoul National University Gwanak Residence Hall | \$1,000  |

## Mentoring

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### Front-Door Project

*Cornell University*

#### RESEARCH MENTOR

*Jan 2026 – August 2026*

- Mentored an undergraduate researcher in computer science department in extending the front-door criterion to more general causal structures through targeted readings and guidance on graphical identification.
- Co-developed generalizable R code for identifying front-door structures and submitted the implementation as a pull request to the developers of the **dagitty** package.

### Harvard Bio Summer Student Research Program

*Harvard University*

#### RESEARCH MENTOR

*June 2024 – August 2024*

- Helped a student translate an interest in medicine and antidepressant use into a researchable causal question using an instrumental-variable framework.
- Guided the student in identifying candidate variables and evaluating possible violations of the required identification assumptions.

### Gwanak Residence Hall

*Seoul National University*

#### RESIDENT ADVISOR

*June 2020 – December 2021*

- Advised undergraduate students on adjusting to university life.
- Planned and organized residence hall festivals.
- Provided assistance to students during COVID-19 isolation.
- Supported international students facing psychological challenges.

### Dream School for Children of Traffic Accident Victims

*Hyundai Motor*

#### DREAM MENTOR

*Spring 2016 – Fall 2016*

- Designed individualized career development programs.
- Mentored a child of traffic accident victims to overcome academic challenges.
- Facilitated opportunities for students to meet inspiring role models.

## Outreach & Professional Development

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### STATISTICAL LANGUAGE

► R, SPSS, Amos,  $\text{\LaTeX}$

### LANGUAGE

► Native: Korean

► Fluent: English

### LICENSURE

► Teacher's Certificate for the Secondary School Teacher of Moral Education, Ministry of Education, Republic of Korea.

## SERVICE

- ▶ Reviewer - CRC Press, Causal Inference Textbook (2026)
- ▶ Reviewer - Journal of Causal Inference (2026)
- ▶ Reviewer - Psychological Methods (2025, 2026)
- ▶ Volunteer - ACIC (American Causal Inference Conference) (2025, 2026)
- ▶ Reviewer - Multivariate Behavioral Research (2025)
- ▶ Reviewer - Scientific Studies of Readings (2025)
- ▶ Reviewer - Psychological Test Adaptation and Development (2025)
- ▶ Student Council in Ewha Womans University (2016-2018)
- ▶ Leader of Educational Academic Society in Ewha Womans University (2016-2017)